Centre for UG and PG Studies (CUPGS), BPUT, Odisha

Department of Electrical Engineering

Power System -II– EEPC3004

6^{th} Sem B Tech & Dual Degree (B. Tech & M. Tech)- Electrical Engineering - 2025-26

Assignment Sheet-1

Date of Issue: 01.03.2026 **Date of Submission: 15.03.2026**

Short type Questions:

1. State two advantages of per unit system.
2. Write an expression for complex power and explain its significance.
3. What do you mean by "Complex Power"?
4. Define Complex Power and draw the power triangle.
5. Write the expression for finding the modified per unit impedance when a new MVA base and kV are chosen.
6. Find the complex power, when the voltage and currents are 100+30j Volt and 25+12j Amp respectively.
7. Name the load flow methods used for transmission system analysis.
8. What is decoupled load flow method?
9. Explain the terms PQ bus and PV bus with reference to load flow analysis in a power system.
10. State the importance of sparsity in power system analysis.
11. State the various types of buses used in load flow analysis of a power system.
12. What is a Heat Rate curve? Briefly explain.
13. What is a Fuel Cost Curve? Briefly explain.
14. What is the expression for generally considered fuel cost of a generator?
15. What is an Incremental Fuel Cost Curve? Briefly explain
16. Briefly, explain Economic Load Dispatch.
17. What are the constraints of economic load dispatch problem?

Long Type

1. A single phase 1000V source is connected across three loads Z_1=60 Ω, Z_2= 6+12j and Z_3=30-30j. Find the power absorbed by each load and the total complex power.
2. Two loads connected in parallel are supplied from a single phase 240V rms source. the two loads draw a total real power of 400kW at a power factor of 0.8 lagging. One of the loads draws120kW at a power factor of 0.96 leading. Find the complex power of the load.
3. Derive the power flow equation for a system containing n-buses and making suitable assumptions.

4. Explain how a bus admittance matrix is modified when a voltage regulator is connected to a particular bus in a power system.
5. Describe the various load flow methods for an interconnected transmission system.
6. Draw a flow chart to explain the Gauss-Seidel method of load flow analysis.
7. Classify various types of buses for power flow studies. Justify the classification.
8. Compare Newton Raphson method with Fast Decoupled method for load flow analysis.
9. Why Newton Raphson method is superior to Gauss Siedel Method. Explain with at least 4 reasons.
10. Critically, compare all the three load flow methods.
11. Compare the three load flow methods: Gauss Siedel, Newton Raphson and Fast Decoupled in terms of storage, execution time and no. of iterations to converge.
12. Explain: Fast Decoupled Load Flow.
13. Explain the Fast Decoupled Load Flow method.
14. State the basic differences between Newton Raphson Load Flow and Fast Decoupled Load Flow method.
15. Modelling of Transformer with off nominal taps.
16. What is a Jacobian matrix? Explain its importance in load flow analysis.
17. What is bus Admittance Matrix? Explain, its significance in Load Flow Analysis.
18. A power system has the impedances between various buses: bus-1 to reference 2j Ω, bus-2 to reference 2j Ω, bus-3 to reference 2j Ω, bus-1 to bus-3 0.2j Ω, bus-2 to bus-3 0.4j Ω, bus-1 to bus-4 0.2j Ω, bus-2 to bus-4 0.4j Ω, bus-3 to bus-4 0.1j Ω. Draw a configuration of the system and find the bus admittance matrix.
19. Explain the economic load dispatch as a constrained optimization problem with and without considering the transmission loss.
20. The fuel cost of 3 thermal plants in Rs/h are given as: $C_1=500+5.3P_1+0.004P_1^2$, $C_2=400+5.5P_2+0.006P_2^2$ and $C_3=200+5.8P_3+0.009P_3^2$, P_1, P_2, P_3 are in MW. The total Load P_D is 800 MW. Neglecting losses and generation limits, find the optimal dispatch and the total cost in Rs/h.
21. The fuel costs of 3 thermal plants in Rs/h are given as: $C_1=500+5.2P_1+0.003P_1^2$,$C_2=400+5.3P_2+0.005P_2^2$, $C_3=200+5.2P_3+0.007P_3^2$, P_1, P_2, P_3 are in MW. The total Load PD is 700 MW. Neglecting losses and generation limits, find the optimal dispatch and total cost in Rs/h.
22. The fuel cost of 3 thermal plants in $/h are given as: $C_1=500+5.3P_1+0.004P_1^2$, $C_2=400+5.5P_2+0.006P_2^2$, $C_3=200+5.8P_3+0.004P_3^2$, P_1, P_2, P_3 are in MW. The total Load PD is 800 MW. Neglecting losses and generation limits, find the optimal dispatch and total cost in $/h using a) analytical method and b) iterative method using gradient search.

Centre for UG and PG Studies (CUPGS), BPUT, Odisha

Department of Electrical Engineering